AUV SIMULATION SCENARIO SPECIFICATIONS

Comprehensive Technical Audit of Actuators, Sensors, Hydrodynamics & Code Placeholders

SCENARIO FILE	SIMULATOR	PHYSICS FRAME	VEHICLE MASS / DEPTH
auv_scenario.xml (v10)	Stonefish ROS 2	NED (+Z Down, SI Units)	156.0 kg / 10.0 m Seabed

8 Thrusters
4 HORIZ VECTORED + 4 VERTICAL

4 Active
SENSORS (IMU, DVL, DEPTH, SONAR)

1 Missing
ZED CAMERA (V10 PLACEHOLDER)

1 Pending
CG OFFSET (SOLIDWORKS CAD)

1 Vehicle Physical & Hydrodynamic Architecture

The simulated vehicle is modeled as a compound submerged rigid body with buoyancy dynamics in North-East-Down (NED) coordinates (**+X Surge / Forward**, **+Y Sway / Starboard**, **+Z Heave / Downward**).

Hull & Rigid Body Variables Dictionary

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & SIGNIFICANCE
`base_link name`	`"Vehicle"`	–	Root dynamic link representing the AUV rigid core to which all actuators and sensors attach.
`base_link physics`	`"submerged"`	–	Enables Stonefish fluid-drag and hydrostatic buoyancy computation for links immersed in water.
`external_part buoyant`	`"true"`	–	Instructs simulator to tetrahedralize collision mesh to calculate submerged volume & Center of Buoyancy (CB).
`physical mesh`	`"auv_hull_phys.obj"`	–	Low-poly watertight 3D mesh used by physics engine for collision checks and volumetric buoyancy integration.
`visual mesh`	`"auv_hull.obj"`	–	High-fidelity 3D mesh rendered visually with PBR materials in the OpenGL simulator window.
`mass value`	`"156"`	kg	Total dry vehicle mass ($156.0\ \text{kg}$). Multiplied by gravity g to determine gravitational downward force ($W \approx 1530.4\ \text{N}$).
`inertia xyz`	`"3.928 3.861 1.164"`	kg·m²	Diagonal moments of inertia: $I_{xx} = 3.928$ (Roll), $I_{yy} = 3.861$ (Pitch), $I_{zz} = 1.164$ (Yaw). Low I_{zz} enables rapid yaw rotation.
`world_transform xyz`	`"0.0 0.0 2.0"`	m	Initial spawn position in NED world frame. $z = 2.0\ \text{m}$ spawns the vehicle $2.0\ \text{m}$ below the surface ($8.0\ \text{m}$ above seabed).

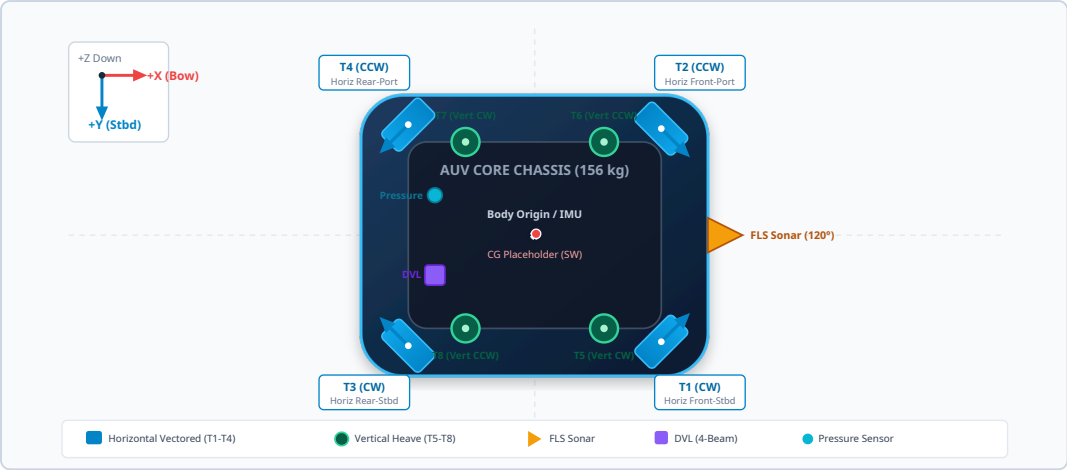
Environment & Ocean Parameters Dictionary

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & SIGNIFICANCE
`ned latitude / longitude`	`"30.0" / "31.0"`	deg	Geographic origin anchor point (Cairo, Egypt). Used by simulated GPS/INS to convert local NED meters to WGS84.
`ocean waves height`	`"0.0"`	m	Surface wave amplitude. Kept at 0.0 to prevent OpenGL compute shader crashes on non-NVIDIA GPUs.
`current velocity`	`"0.0 0.0 0.0"`	m/s	Uniform ocean water current vector $[u_x, u_y, u_z]$. Currently quiescent; can be adjusted for disturbance rejection tests.
`static Bottom transform`	`"0.0 0.0 10.0"`	m	Horizontal rock seabed plane located at $z = 10.0\ \text{m}$ depth. Defines bottom altitude boundary for DVL tracking.

CRITICAL CODE PLACEHOLDER: Center of Gravity (CG) Offset Commented Out
In line 64 of `auv_scenario.xml`: `<!-- <cg xyz="0.00281 -0.00154 -0.3092" rpy="0 0 0"/> -->`.
What it means: Omitting `<cg>` sets the Center of Mass (CM) at geometric origin $(0,0,0)$. The CAD value has $z = -0.3092\ \text{m}$ (above origin in NED). If the Center of Buoyancy (CB) is below this point, metacentric height $\overline{GM} < 0$, causing capsizing. Verify ballast before uncommenting.

2 8-Thruster & Sensor Vector Geometry (Top View)

The AUV employs an 8-thruster arrangement combining four horizontal diagonal vectored thrusters (T1–T4) with four vertical heave thrusters (T5–T8) in the top plate corner holes.



3 Thruster / Actuator Variables Master Dictionary

The table below explains every single parameter defined inside the `<actuator type="thruster">` XML blocks and its physical effect on simulation dynamics:

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & PHYSICAL EFFECT
`actuator name`	`"ThrusterT1" .. T8`	–	Unique identifier for the actuator within the robot kinematic tree. Used by ROS 2 setpoint dispatcher.
`specs max_setpoint`	`"100.0"`	rad/s	Maximum rotational angular velocity of the propeller ($\omega_{\text{max}} = 100\ \text{rad/s} \approx 955\ \text{RPM}$).
`specs normalized_setpoint`	`"true"`	boolean	When true, ROS command input values are expected in normalized range $[-1.0, 1.0]$. Input 1.0 yields ω_{max}.
`specs inverted_setpoint`	`"false"`	boolean	If true, inverts the algebraic sign of commanded setpoint. Kept false since chirality handles direction.
`propeller diameter`	`"0.1"`	m	Propeller blade tip diameter ($D = 10.0\ \text{cm}$). Used for disc area calculations and mesh scaling.
`propeller right`	`"true" / "false"`	boolean	Chirality / handedness: `true` = Right-handed (CW), `false` = Left-handed (CCW). Crucial for torque balancing.
`rotor_dynamics time_constant`	`"0.1"`	s	First-order lag $\tau = 0.1\ \text{s}$. Propeller reaches 63.2% of commanded RPM in $100\ \text{ms}$, modeling motor inertia.
`thrust_coeff forward`	`"0.88"`	N/(rad/s)²	Forward thrust factor C_T in $T = C_T \cdot \omega \|\omega\|$. Generates $0.88\ \text{N}$ at $\omega = 1\ \text{rad/s}$.
`thrust_coeff reverse`	`"0.88"`	N/(rad/s)²	Reverse thrust factor. Set symmetric to forward in current XML (placeholder for real propeller asymmetry).
`torque_coeff value`	`"0.05"`	N·m/(rad/s)²	Hydrodynamic drag counter-torque $Q = C_Q \cdot \omega \|\omega\|$ transmitted back into the AUV body around shaft axis.
`origin xyz / rpy`	`(varies per thruster)`	m / rad	Mounting 3D offset and rotation. Thrust vector is directed along the local thruster $+X$ axis.

3.1 Individual Thruster Parameters Breakdown (T1 to T8)

Detailed positional and angular parameters for all eight thrusters. Roll, pitch, and yaw angles are expressed in both radians and degrees:

ID	PHYSICAL ROLE & LOCATION	MOUNTING ORIGIN [M]	ORIENTATION (RPY)	CHIRALITY	THRUST VECTOR DIRECTION	PRIMARY DOF CONTROLLED
T1	Horizontal Front-Starboard	`X: +0.2537` `Y: +0.2213` `Z: +0.0227`	`R: 0.0 rad (0°)` `P: 0.0 rad (0°)` `Y: -0.7854 rad (-45°)`	**RIGHT (CW)**	$[+0.7071, -0.7071, 0.0]$ Forward & Port push	Surge ($+F_x$), Sway ($-F_y$), Yaw ($-M_z$)
T2	Horizontal Front-Port	`X: +0.2537` `Y: -0.2213` `Z: +0.0227`	`R: 0.0 rad (0°)` `P: 0.0 rad (0°)` `Y: +0.7854 rad (+45°)`	**LEFT (CCW)**	$[+0.7071, +0.7071, 0.0]$ Forward & Stbd push	Surge ($+F_x$), Sway ($+F_y$), Yaw ($+M_z$)
T3	Horizontal Rear-Starboard	`X: -0.2463` `Y: +0.2287` `Z: +0.0227`	`R: 0.0 rad (0°)` `P: 0.0 rad (0°)` `Y: +3.9270 rad (+225°)`	**RIGHT (CW)**	$[-0.7071, -0.7071, 0.0]$ Aft & Port push	Surge ($-F_x$), Sway ($-F_y$), Yaw ($-M_z$)
T4	Horizontal Rear-Port	`X: -0.2463` `Y: -0.2287` `Z: +0.0227`	`R: 0.0 rad (0°)` `P: 0.0 rad (0°)` `Y: -3.9270 rad (-225°)`	**LEFT (CCW)**	$[-0.7071, +0.7071, 0.0]$ Aft & Stbd push	Surge ($-F_x$), Sway ($+F_y$), Yaw ($+M_z$)
T5	Vertical Front-Starboard	`X: +0.1430` `Y: +0.1945` `Z: -0.0565`	`R: 0.0 rad (0°)` `P: -1.5708 rad (-90°)` `Y: 0.0 rad (0°)`	**RIGHT (CW)**	$[0.0, 0.0, +1.0]$ Pushes Downward (+Z)	Heave ($+F_z$), Pitch ($-M_y$), Roll ($+M_x$)
T6	Vertical Front-Port	`X: +0.1430` `Y: -0.1945` `Z: -0.0565`	`R: 0.0 rad (0°)` `P: +1.5708 rad (+90°)` `Y: 0.0 rad (0°)`	**LEFT (CCW)**	$[0.0, 0.0, -1.0]$ Pushes Upward (-Z)	Heave ($-F_z$), Pitch ($+M_y$), Roll ($+M_x$)
T7	Vertical Rear-Port	`X: -0.1430` `Y: -0.1945` `Z: -0.0565`	`R: 0.0 rad (0°)` `P: -1.5708 rad (-90°)` `Y: 0.0 rad (0°)`	**RIGHT (CW)**	$[0.0, 0.0, +1.0]$ Pushes Downward (+Z)	Heave ($+F_z$), Pitch ($+M_y$), Roll ($-M_x$)
T8	Vertical Rear-Starboard	`X: -0.1430` `Y: +0.1945` `Z: -0.0565`	`R: 0.0 rad (0°)` `P: +1.5708 rad (+90°)` `Y: 0.0 rad (0°)`	**LEFT (CCW)**	$[0.0, 0.0, -1.0]$ Pushes Upward (-Z)	Heave ($-F_z$), Pitch ($-M_y$), Roll ($-M_x$)

Motor Lag & Thrust Mathematical Equations

1st-Order Electro-Mechanical Lag:
$$\tau \frac{d\omega}{dt} + \omega = \omega_{\text{cmd}},\quad \text{with } \tau = 0.10\ \text{s}$$ **Hydrodynamic Thrust & Reaction Torque Law:**
$$T = C_T \cdot \omega |\omega| = 0.88 \cdot \omega |\omega|\ [\text{N}]$$
$$Q = C_Q \cdot \omega |\omega| = 0.05 \cdot \omega |\omega|\ [\text{N}\cdot\text{m}]$$

ROS 2 Command Subscription Interface

Subscriber Topic: `/auv/thruster_setpoints`
Message Type: `std_msgs/msg/Float64MultiArray`
Array Format: Float vector with 8 elements: $[u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_8]$.
Input Range: $[-1.0, 1.0]$. The value 0.0 stops rotation; 1.0 spins at $\omega_{\text{max}} = 100\ \text{rad/s}$.

Thrust Allocation Matrix (TAM) Mapping Matrix B

Maps the 6-DOF force/torque vector $\mathbf{\tau} = [F_x, F_y, F_z, M_x, M_y, M_z]^T$ to the 8 thruster setpoints ($\mathbf{\tau} = \mathbf{B} \cdot \mathbf{u}$):

```
# Rows:    [ Surge (Fx), Sway (Fy), Heave (Fz), Roll (Mx), Pitch (My), Yaw (Mz) ]
# Columns: [ T1,         T2,        T3,         T4,        T5,         T6,        T7,         T8 ]
[  0.7071,  0.7071, -0.7071, -0.7071,     0.0,       0.0,       0.0,       0.0   ]  # Fx (Surge)
[ -0.7071,  0.7071, -0.7071,  0.7071,     0.0,       0.0,       0.0,       0.0   ]  # Fy (Sway)
[     0.0,     0.0,     0.0,     0.0,  1.0000,   -1.0000,    1.0000,   -1.0000   ]  # Fz (Heave)
[     0.0,     0.0,     0.0,     0.0,  0.1945,    0.1945,   -0.1945,   -0.1945   ]  # Roll (Mx)
[     0.0,     0.0,     0.0,     0.0, -0.1430,    0.1430,    0.1430,   -0.1430   ]  # Pitch (My)
[ -0.3359,  0.3359, -0.3359,  0.3359,     0.0,       0.0,       0.0,       0.0   ]  # Yaw (Mz)
```

4 IMU & DVL Sensor Variables Deep-Dive

Exhaustive parameter breakdown for the Inertial Measurement Unit (IMU) and Doppler Velocity Log (DVL):

Sensor 1: Inertial Measurement Unit (<sensor name="IMU" type="imu">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"50.0"`	Hz	Output frequency. IMU publishes 50 messages per second (every 20 ms), matching requirements for high-bandwidth attitude filters (EKF/Madgwick).
`range angular_velocity`	`"10.0 10.0 10.0"`	rad/s	Full-scale gyroscope measurement ceiling along X, Y, Z axes ($\pm 10.0\ \text{rad/s} \approx \pm 573^\circ/\text{s}$). Exceeding this value causes digital saturation.
`range linear_acceleration`	`"20.0"`	m/s²	Full-scale accelerometer measurement ceiling ($\pm 20.0\ \text{m/s}^2 \approx \pm 2.04\ \text{g}$). Accurately captures gravitational vector plus linear vehicle accelerations.
`noise angle`	`"0.001 0.001 0.001"`	rad	White Gaussian noise standard deviation added to the orientation quaternion ($1.0\ \text{mrad} \approx 0.057^\circ$). Simulates AHRS attitude error.
`noise angular_velocity`	`"0.0001"`	rad/s	Gyroscope measurement noise standard deviation ($0.0001\ \text{rad/s} \approx 0.0057^\circ/\text{s}$). Simulates tactical-grade MEMS gyroscope noise density.
`noise yaw_drift`	`"0.0001"`	rad/s	Continuous uncompensated heading drift rate. Simulates gyro bias instability (unbounded random walk) in the absence of a magnetic compass aiding.
`noise linear_acceleration`	`"0.001"`	m/s²	Accelerometer measurement white noise standard deviation ($0.001\ \text{m/s}^2 \approx 0.1\ \text{mg}$). Represents high-precision accelerometer output.
`history samples`	`"1"`	samples	Internal filter queue depth. 1 sample means raw instantaneous readings are published with zero moving-average group delay.
`origin xyz / rpy`	`"0.0 0.0 0.0" / "0 0 0"`	m / rad	Mounted exactly at body geometric center. Eliminates parasitic centrifugal and tangential acceleration offsets $(\mathbf{\omega} \times (\mathbf{\omega} \times \mathbf{r}))$.
`ros_publisher topic`	`"/auv/imu"`	–	Target ROS 2 topic emitting standard message type `sensor_msgs/msg/Imu`.

Sensor 2: Doppler Velocity Log (<sensor name="DVL" type="dvl">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"5.0"`	Hz	Acoustic ping update frequency. Emits 5 velocity packets per second (every 200 ms), typical for subsea acoustic travel time over depth.
`specs beam_angle`	`"30.0"`	deg	Elevation tilt angle of the 4 Janus transducer beams relative to vertical axis (30.0°). Used by Janus transformation matrix to resolve 3D body velocity.
`range velocity`	`"3.0 3.0 1.0"`	m/s	Maximum measurable speed-over-ground: $\pm 3.0\ \text{m/s}$ in Surge (X), $\pm 3.0\ \text{m/s}$ in Sway (Y), and $\pm 1.0\ \text{m/s}$ in Heave (Z).
`range altitude_min / max`	`"0.5" / "50.0"`	m	Operational bottom-lock window: minimum blanking zone of $0.5\ \text{m}$ (near-field transducer ringing) up to $50.0\ \text{m}$ maximum acoustic bottom track.
`noise velocity`	`"0.0015"`	m/s	Velocity measurement noise standard deviation ($1.5\ \text{mm/s}$ RMS). Provides extremely high-accuracy velocity aiding for dead reckoning.
`noise altitude`	`"0.001"`	m	Acoustic altimeter range noise standard deviation ($1.0\ \text{mm}$ RMS). Precision distance to seabed.
`history samples`	`"1"`	samples	No internal sliding window averaging applied; publishes single-ping instantaneous Doppler shifts.
`origin xyz / rpy`	`"-0.19 0.19 0.45" / "0 0 0"`	m / rad	Mounted on aft-starboard underside ($x=-0.19\text{m}$, $y=+0.19\text{m}$, $z=+0.45\text{m}$), pointing directly downwards into water column.
`ros_publisher topic`	`"/auv/dvl"`	–	Publishes `stonefish_ros2/msg/DVL` (velocity vector, beam ranges, validity flags). Optional altitude published to `sensor_msgs/Range`.

4.1 Pressure Sensor, FLS Sonar & Camera Variables Deep-Dive

Exhaustive parameter breakdown for the Subsea Pressure Sensor, Forward-Looking Multibeam Sonar (FLS), and ZED Camera:

Sensor 3: Hydrostatic Depth / Pressure (<sensor name="Pressure" type="pressure">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"10.0"`	Hz	Output frequency. Pressure readings emitted 10 times per second (every 100 ms) for closed-loop depth holding.
`range pressure`	`"1000000.0"`	Pa	Maximum pressure limit ($10^6\ \text{Pa} = 10\ \text{bar}$). Derived depth: $z \approx \frac{P - P_0}{\rho g} \approx \frac{9\times 10^5}{1000 \times 9.81} \approx 91.7\ \text{m}$ subsea rating.
`noise pressure`	`"10.0"`	Pa	Hydrostatic pressure noise standard deviation ($10.0\ \text{Pa}$). Equivalent to millimeter depth resolution: $\Delta z = \frac{10}{1000 \times 9.81} \approx 1.02\ \text{mm}\ \text{H}_2\text{O}$.
`history samples`	`"1"`	samples	Single sample history without lag.
`origin xyz / rpy`	`"-0.19 0.19 0.20" / "0 0 0"`	m / rad	Port tap position located $0.20\ \text{m}$ below body geometric center ($x=-0.19\text{m}$, $y=+0.19\text{m}$).
`ros_publisher topic`	`"/auv/pressure"`	–	Target ROS 2 topic emitting standard message type `sensor_msgs/msg/FluidPressure`.

Sensor 4: Forward-Looking Imaging Sonar (<sensor name="FLS" type="fls">)

PARAMETER / XML ATTRIBUTE	VALUE IN CODE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"10.0"`	Hz	Acoustic frame rate. Generates 10 complete 2D acoustic echogram images per second.
`specs beams`	`"512"`	beams	Number of horizontal acoustic receiver beams across the aperture. Yields angular resolution $\Delta \theta = \frac{120^\circ}{512} \approx 0.234^\circ$ per beam.
`specs bins`	`"500"`	bins	Number of radial range sample cells per beam. Yields range resolution $\Delta r = \frac{10.0 - 0.5}{500} = 1.9\ \text{cm}$ per cell.
`specs horizontal_fov`	`"120.0"`	deg	Total horizontal azimuthal aperture opening angle (120.0°). Provides wide-angle forward situational awareness.
`specs vertical_fov`	`"120.0"`	deg	Vertical beam width cone (120.0°). Ensures acoustic illumination from water surface down to bottom.
`specs output_format`	`"uint8"`	–	8-bit unsigned integer per pixel ($0 - 255$). 0 = no acoustic backscatter (clear water), 255 = maximum specular echo return.
`settings range_min / max`	`"0.5" / "10.0"`	m	Active acoustic imaging range: $0.5\ \text{m}$ near-field blind zone up to $10.0\ \text{m}$ maximum obstacle detection distance.
`settings gain`	`"1.1"`	–	Acoustic receiver amplification factor ($1.1\times$). Boosts weak echoes from distant or low-reflectivity targets.
`noise multiplicative / add`	`"0.01" / "0.02"`	–	Simulates acoustic clutter: 1% multiplicative speckle noise plus 2% additive thermal/ambient ambient water noise.
`display colormap`	`"perula"`	–	Color lookup table for Stonefish visualizer GUI rendering (blue-green-yellow high-contrast palette).
`origin xyz / rpy`	`"0.35 0.0 -0.10" / "0 1.5708 0"`	m / rad	Mounted at bow ($x=+0.35\text{m}$, $z=-0.10\text{m}$) pitched by 90° ($1.5708\ \text{rad}$) to aim acoustic cone directly ahead.
`ros_publisher topic`	`"/auv/display"`	–	Publishes raw 2D polar echogram matrix as a standard ROS 2 `sensor_msgs/msg/Image`.

Sensor 5: ZED 2i Stereo Camera (Planned / Header Placeholder)

PARAMETER / XML ATTRIBUTE	RECOMMENDED VALUE	UNIT	DETAILED ENGINEERING MEANING & WHAT IT DOES
`rate`	`"30.0"`	Hz	Real-time video frame rate (30 FPS) for visual odometry (VIO) and YOLO deep learning object detection.
`specs resolution`	`1280 x 720`	pixels	HD 720p color video stream providing high feature density without exceeding GPU rendering bandwidth.
`specs horizontal_fov`	`"110.0"`	deg	Wide field-of-view matching the physical Stereolabs ZED 2i optical lens underwater profile.
`origin xyz / rpy`	`"0.32 0.0 0.05" / "0`	m / rad	Bow forward mount ($x=+0.32\text{m}$, deck level $z=+0.05\text{m}$)

5 Code Placeholders & Architectural Recommendations

1. MISSING HARDWARE SENSOR: ZED Stereo Camera Tag
Audit Finding: Line 5 header declares: `v10: + DVL, pressure, 1D sonar, ZED.` However, no camera sensor tag exists in lines 216–255! Vision pipelines (YOLO gate/buoy detection, VIO, visual SLAM) cannot launch without camera topics.

2. UNENABLED TELEMETRY: Thruster State Feedback Publisher
Audit Finding: Line 256 defines `<ros_subscriber thrusters="/auv/thruster_setpoints"/>`, but lacks thruster state telemetry feedback.
Enhancement: Adding `<ros_publisher thrusters="/auv/thruster_state"/>` exposes real-time propeller RPM, actual thrust force, and torque feedback.

3. GPU CRASH SAFEGUARD: Ocean Wave Height Locked at 0.0
Technical Context: Line 14 sets `<waves height="0.0"/>` with the comment `Keep 0.0: any wave height crashes the shaders on your GPU.` Stonefish wave surface rendering uses OpenGL compute shaders. Keep 0.0 when running on integrated GPUs or software Mesa drivers.

4. OCEAN CURRENT DISTURBANCE & SENSOR SERVICE CONTROLS
• **Current Velocity:** `<velocity xyz="0.0 0.0 0.0"/>` is currently quiescent. Inject non-zero values (e.g. 0.25 m/s) to benchmark disturbance rejection.
• **Dynamic Power Services:** Add `<ros_service set_enabled="/auv/sensor_name/set_enabled"/>` to toggle heavy sensors like the 512-beam FLS sonar via `std_srvs/srv/SetBool`.

6 Ready-to-Use XML Drop-in Code Snippets

A. ZED 2i Stereo Camera (Insert before <ros_subscriber>)

```
<!-- ================= ZED 2i STEREO CAMERA ================= -->
<sensor name="ZED_Camera" rate="30.0" type="camera">
    <specs resolution_x="1280" resolution_y="720" horizontal_fov="110.0"/>
    <origin xyz="0.32 0.0 0.05" rpy="0.0 0.0 0.0"/>
    <link name="Vehicle"/>
    <ros_publisher topic="/auv/zed/image_raw"/>
    <ros_service set_enabled="/auv/zed/set_enabled"/>
</sensor>
```

B. Thruster State Feedback & Verified Center of Gravity

```
<!-- Enable real-time telemetry of thruster RPM, thrust force, and torque -->
<ros_publisher thrusters="/auv/thruster_state"/>

<!-- Verified SolidWorks CG (Uncomment after checking ballast stability GM > 0) -->
<cg xyz="0.00281 -0.00154 -0.3092" rpy="0.0 0.0 0.0"/>
```

7 Master ROS 2 Topics & Interfaces Summary

TOPIC NAME	DIRECTION	MESSAGE TYPE	RATE	DESCRIPTION
`/auv/thruster_setpoints`	SUB	`std_msgs/msg/Float64MultiArray`	Async	8-element vector of normalized thruster setpoints [-1.0, 1.0]
`/auv/thruster_state`	PUB	`stonefish_ros2/msg/ThrusterState`	Sync	Real-time feedback: setpoints, angular velocities (rad/s), thrusts (N)
`/auv/imu`	PUB	`sensor_msgs/msg/Imu`	50 Hz	Linear accelerations, angular rates, and fused orientation quaternion
`/auv/dvl`	PUB	`stonefish_ros2/msg/DVL`	5 Hz	3-axis bottom-track velocity (m/s) and beam altitude measurements
`/auv/pressure`	PUB	`sensor_msgs/msg/FluidPressure`	10 Hz	Hydrostatic water pressure (Pa) for precise depth estimation
`/auv/display`	PUB	`sensor_msgs/msg/Image`	10 Hz	Forward-looking sonar polar acoustic intensity image (uint8)